S.No: 4	Exp. Name: ***Generate Pay Slips for Employees***	Date:2026-10-17

ID: 8115U25EC001

2025-2029-ECE-A

K.Ramakrishnan College of Engineering

Aim:

You are tasked with developing a Java application to manage salary information for employees in an organization. The application should include the following classes:

1. **Employee** class with the following attributes:
   - empName (Employee Name)
   - empId (Employee ID)
   - address (Employee Address)
   - mailId (Employee Email ID)
   - mobileNo (Employee Mobile Number)

2. **Subclasses** that inherit from the **Employee** class:
   - Programmer
   - AssistantProfessor
   - AssociateProfessor
   - Professor

Each of these subclasses should also include an additional attribute called **basicPay**.

For all the employee types (subclasses), you need to calculate the following components of their salary:

1. Dearness Allowance (**DA**): 97% of the basicPay
2. House Rent Allowance (**HRA**): 10% of the basicPay
3. Provident Fund (**PF**): 12% of the basicPay
4. Staff Club Fund: 0.1% of the basicPay

Additionally, you should compute and display the following salary details for each employee:

1. **Gross Salary**: Sum of basicPay, DA, and HRA
2. **Net Salary**: Gross Salary minus PF and Staff Club Fund

Your application should allow the user to input the details of employees, including their name, ID, address, email ID, mobile number, and basic pay. After inputting these details, the program should generate and display a pay slip for each employee, including all the salary components and the final gross and net salaries.

Instructions:

1. Create the necessary classes (`Employee`, `Programmer`, `AssistantProfessor`, `AssociateProfessor`, `Professor`) with appropriate attributes and methods.
2. Implement a method in each subclass to calculate the salary components and display the pay slip.
3. In the `SalaryManagementSystem` class, take user input for each type of employee, create instances of the corresponding subclass, and generate pay slips for each employee.
4. Display all the relevant information for each employee, including basic pay, DA, HRA, PF, Staff Club Fund, Gross Salary, and Net Salary.

Source Code:

SalaryManagementSystem.java

```
import java.util.Scanner;

class Employee {
        protected String empName;
```

```
    protected String empId;
    protected String address;
    protected String mailId;
    protected String mobileNo;

    public Employee(String empName, String empId, String address, String mailId, Stri
ng mobileNo) {
        this.empName = empName;
        this.empId = empId;
        this.address = address;
        this.mailId = mailId;
        this.mobileNo = mobileNo;
    }

    public void displayBasicInfo() {
        System.out.println("Pay Slip for " + empName);
        System.out.println("Employee ID: " + empId);
        System.out.println("Address: " + address);
        System.out.println("Mail ID: " + mailId);
        System.out.println("Mobile Number: " + mobileNo);
        }// write your code here...


}

class Programmer extends Employee{
        private double basicPay;

    public Programmer(String empName, String empId, String address, String mailId, St
ring mobileNo, double basicPay) {
        super(empName, empId, address, mailId, mobileNo);
        this.basicPay = basicPay;
    }

    public void generatePaySlip() {
        super.displayBasicInfo();
        double da = 0.97 * basicPay;
        double hra = 0.10 * basicPay;
        double pf = 0.12 * basicPay;
        double staffClubFund = 0.001 * basicPay;
        double grossSalary = basicPay + da + hra;
        double netSalary = grossSalary - (pf + staffClubFund);

        System.out.println("Basic Pay: " + basicPay);
        System.out.println("DA: " + da);
        System.out.println("HRA: " + hra);
        System.out.println("PF: " + pf);
        System.out.println("Staff Club Fund: " + staffClubFund);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("Net Salary: " + netSalary);
        }//write your code here...


}

class AssistantProfessor extends Employee{
```

```
        private double basicPay;

    public AssistantProfessor(String empName, String empId, String address, String ma
ilId, String mobileNo, double basicPay) {
        super(empName, empId, address, mailId, mobileNo);
        this.basicPay = basicPay;
    }

    public void generatePaySlip() {
        super.displayBasicInfo();
        double da = 0.97 * basicPay;
        double hra = 0.10 * basicPay;
        double pf = 0.12 * basicPay;
        double staffClubFund = 0.001 * basicPay;
        double grossSalary = basicPay + da + hra;
        double netSalary = grossSalary - (pf + staffClubFund);

        System.out.println("Basic Pay: " + basicPay);
        System.out.println("DA: " + da);
        System.out.println("HRA: " + hra);
        System.out.println("PF: " + pf);
        System.out.println("Staff Club Fund: " + staffClubFund);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("Net Salary: " + netSalary);
        }//write your code here...


}

class AssociateProfessor extends Employee{
        private double basicPay;

    public AssociateProfessor(String empName, String empId, String address, String ma
ilId, String mobileNo, double basicPay) {
        super(empName, empId, address, mailId, mobileNo);
        this.basicPay = basicPay;
    }

    public void generatePaySlip() {
        super.displayBasicInfo();
        double da = 0.97 * basicPay;
        double hra = 0.10 * basicPay;
        double pf = 0.12 * basicPay;
        double staffClubFund = 0.001 * basicPay;
        double grossSalary = basicPay + da + hra;
        double netSalary = grossSalary - (pf + staffClubFund);

        System.out.println("Basic Pay: " + basicPay);
        System.out.println("DA: " + da);
        System.out.println("HRA: " + hra);
        System.out.println("PF: " + pf);
        System.out.println("Staff Club Fund: " + staffClubFund);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("Net Salary: " + netSalary);
        }//write your code here...
```

```
}

class Professor extends Employee{
        private double basicPay;

    public Professor(String empName, String empId, String address, String mailId, Str
ing mobileNo, double basicPay) {
        super(empName, empId, address, mailId, mobileNo);
        this.basicPay = basicPay;
    }

    public void generatePaySlip() {
        super.displayBasicInfo();
        double da = 0.97 * basicPay;
        double hra = 0.10 * basicPay;
        double pf = 0.12 * basicPay;
        double staffClubFund = 0.001 * basicPay;
        double grossSalary = basicPay + da + hra;
        double netSalary = grossSalary - (pf + staffClubFund);

        System.out.println("Basic Pay: " + basicPay);
        System.out.println("DA: " + da);
        System.out.println("HRA: " + hra);
        System.out.println("PF: " + pf);
        System.out.println("Staff Club Fund: " + staffClubFund);
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("Net Salary: " + netSalary);
        }//write your code here...


}

public class SalaryManagementSystem {
    public static void main(String[] args) {
                Scanner input = new Scanner(System.in);

        System.out.println("Enter details for Programmer:");
        System.out.print("Employee Name: ");
        String name = input.nextLine();
        System.out.print("Employee ID: ");
        int id = input.nextInt();
        input.nextLine();
        System.out.print("Address: ");
        String address = input.nextLine();
        System.out.print("Mail ID: ");
        String mailId = input.nextLine();
        System.out.print("Mobile Number: ");
        long mobileNo = input.nextLong();
        System.out.print("Basic Pay: ");
        double basicPay = input.nextDouble();

        Programmer programmer = new Programmer(name, String.valueOf(id), address, mai
lId, String.valueOf(mobileNo), basicPay);
        programmer.generatePaySlip();// create an object of Programmer class and call
by passing the above inputs as parameters
```


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```
        System.out.println("Enter details for Assistant Professor:");
        input.nextLine();
        System.out.print("Employee Name: ");
        name = input.nextLine();
        System.out.print("Employee ID: ");
        id = input.nextInt();
        input.nextLine();
        System.out.print("Address: ");
        address = input.nextLine();
        System.out.print("Mail ID: ");
        mailId = input.nextLine();
        System.out.print("Mobile Number: ");
        mobileNo = input.nextLong();
        System.out.print("Basic Pay: ");
        basicPay = input.nextDouble();

        AssistantProfessor assistantProfessor = new AssistantProfessor(name, String.v
alueOf(id), address, mailId, String.valueOf(mobileNo), basicPay);
        assistantProfessor.generatePaySlip();// create an object of AssistantProfesso
r class and call it with the input values...



        System.out.println("Enter details for Associate Professor:");
        input.nextLine();
        System.out.print("Employee Name: ");
        name = input.nextLine();
        System.out.print("Employee ID: ");
        id = input.nextInt();
        input.nextLine();
        System.out.print("Address: ");
        address = input.nextLine();
        System.out.print("Mail ID: ");
        mailId = input.nextLine();
        System.out.print("Mobile Number: ");
        mobileNo = input.nextLong();
        System.out.print("Basic Pay: ");
        basicPay = input.nextDouble();

                // create an object of AssociateProfessor class and call it with the
input values...
        AssociateProfessor associateProfessor = new AssociateProfessor(name, String.v
alueOf(id), address, mailId, String.valueOf(mobileNo), basicPay);
        associateProfessor.generatePaySlip();



        System.out.println("Enter details for Professor:");
        input.nextLine();
        System.out.print("Employee Name: ");
        name = input.nextLine();
        System.out.print("Employee ID: ");
        id = input.nextInt();
        input.nextLine();
```

```
        System.out.print("Address: ");
        address = input.nextLine();
        System.out.print("Mail ID: ");
        mailId = input.nextLine();
        System.out.print("Mobile Number: ");
        mobileNo = input.nextLong();
        System.out.print("Basic Pay: ");
        basicPay = input.nextDouble();

        Professor professor = new Professor(name, String.valueOf(id), address, mailI
d, String.valueOf(mobileNo), basicPay);
        professor.generatePaySlip();
        // create an object of Professor class and call it with the input values...


        input.close();
    }
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter details for Programmer:
Employee Name:
John
Employee ID:
100
Address:
Princeton
Mail ID:
j23@gmail.com
Mobile Number:
9524304564
Basic Pay:
10000
Pay Slip for John
Employee ID: 100
Address: Princeton
Mail ID: j23@gmail.com
Mobile Number: 9524304564
Basic Pay: 10000.0
DA: 9700.0
HRA: 1000.0
PF: 1200.0
Staff Club Fund: 10.0
Gross Salary: 20700.0
Net Salary: 19490.0
Enter details for Assistant Professor:
Employee Name:

Clarie
Employee ID:
101
Address:
Ottawa
Mail ID:
cl34@gmail.com
Mobile Number:
8500212274
Basic Pay:
12000
Pay Slip for Clarie
Employee ID: 101
Address: Ottawa
Mail ID: cl34@gmail.com
Mobile Number: 8500212274
Basic Pay: 12000.0
DA: 11640.0
HRA: 1200.0
PF: 1440.0
Staff Club Fund: 12.0
Gross Salary: 24840.0
Net Salary: 23388.0
Enter details for Associate Professor:
Employee Name:
Licoln
Employee ID:
103
Address:
Washington
Mail ID:
linc23@gmail.com
Mobile Number:
9100390742
Basic Pay:
14000
Pay Slip for Licoln
Employee ID: 103
Address: Washington
Mail ID: linc23@gmail.com
Mobile Number: 9100390742
Basic Pay: 14000.0
DA: 13580.0
HRA: 1400.0
PF: 1680.0
Staff Club Fund: 14.0
Gross Salary: 28980.0
Net Salary: 27286.0
Enter details for Professor:
Employee Name:

Krem
Employee ID:
105
Address:
Loss angels
Mail ID:
krim@gmail.com
Mobile Number:
9010692094
Basic Pay:
15000
Pay Slip for Krem
Employee ID: 105
Address: Loss angels
Mail ID: krim@gmail.com
Mobile Number: 9010692094
Basic Pay: 15000.0
DA: 14550.0
HRA: 1500.0
PF: 1800.0
Staff Club Fund: 15.0
Gross Salary: 31050.0
Net Salary: 29235.0